COE548/748 Final Project: Building a Specialized LLM Agent

Project Overview

This project involves designing and developing a specialized Large Language Model (LLM) agent. You may use any modern LLM framework (e.g., LangChain, LlamaIndex, or others) and any model provider (e.g., Gemini, OpenAI, open-source models, etc.). The goal is to create a useful AI-powered system that solves a meaningful problem. Your agent should integrate retrieval augmented generation (RAG) and custom tools, and should be accessible through a user interface (e.g., Streamlit, Gradio, or web app). You have the freedom to determine the primary function and use-case of your LLM agent.

Students will work in groups of three.

Objectives

1. Develop a custom LLM agent capable of solving a specialized task.
2. Integrate multiple tools including Retrieval-Augmented Generation (RAG).
3. Design a user interface for interacting with the agent.
4. Demonstrate proper API and data integration.
5. Follow good software engineering practices.
6. Document system design, implementation decisions, and experimental results.

Core Implementation Requirements

- Create a Custom LLM Agent using any LLM framework of your choice.
- Implement Retrieval Augmented Generation (RAG) using vector embeddings.
- Integrate at least three additional tools (APIs, external services, etc.), one of which must be a customized tool you design.
- Develop an interface for interacting with the agent (Streamlit, Gradio, web app, etc.).
- Maintain conversation history and implement proper error handling.

Tools and Technologies

There is no restriction on the models, providers, API, libraries, or technologies that you use. You are encouraged to experiment and integrate any technologies that improve the quality or capabilities of your system.

Submission Requirements:

1- Report
2- Code
3- 7 minutes Presentation including a video demo

Final Report Requirements (IEEE Format)

The final report must be written in IEEE conference paper format and must be no more than 6 pages INCLUDING references.

You must use the official IEEE Word template available here:
https://www.ieee.org/conferences/publishing/templates.html

Literature Review Requirement

Your report must include a literature review referencing at least 10 research papers published between 2025 and 2026. These references must be properly cited using IEEE citation style.

In this section, you must clearly explain how your system differs from existing work. This section should highlight the novelty, creativity, and unique contributions of your project compared to the referenced literature.

Required Sections in the Report

- Abstract
- Introduction
- Related Work (Literature Review)
- Methodology
    - High level overview of the system architecture and components
    - System design diagram
    - Tool documentation
    - …
- Experimental Setup
    - Models
    - Datasets
    - Prompts and Design Decisions
    - Evaluation Metrics
    - …
- Results and Discussion
- Conclusion and Future Work
- Author Contributions (list who did what in the project)
- References

Code Submission

You must submit the complete project code along with your report. The repository or folder should include all source code, configuration files, etc., and a requirements.txt (or equivalent) file listing dependencies. Include instructions on how to run the code in a README.txt file.

Presentation and Demo

Students must submit and deliver a 7-minute presentation that includes a video demonstration of the system. The demonstration should clearly showcase the main functionalities of the project and explain its key features while the system is in operation.

Evaluation Criteria

- Functionality (30%) – Does the system work as intended? Are all core and additional requirements met?
- Report Quality (30%) – Is the IEEE report clear, well-structured, and complete?
- Code Quality (15%) – Is the code readable, modular, and well-documented?
- Presentation (15%) – Is the presentation clear? Is the demo clear and effective? Does it highlight the main features and functionalities?
- Innovation and Creativity (10%) – Does the project demonstrate originality and meaningful improvements over related work?

Important Notes

- Do NOT include API keys in submitted files (for your own sake)
- Use placeholders such as [INSERT API KEY HERE].
- Store real API keys securely using environment variables (.env files recommended).
- Respect licenses of all third-party tools used.
- Ensure any data used respects privacy and ethical standards.